It's important to notice here that the circuit in the breadboard does not necessarily look like the schematic. That's because schematics are not meant to be a geographical map of what the circuit looks like but rather they simply show the connections that need to be made. Connecting one pin of the LED and one pin of the resistor into any of the five holes that share a connection will connect the LED to the resistor, completing that part of the circuit. Starting at the top left-hand side of the schematic, we see that the positive side of the battery connects to one side of the switch, the other side of the switch connects to the resistor, the second leg of the resistor connects to the positive or anode pin of the LED, and the negative pin or cathode of the LED connects to the negative terminal of the battery. We could place the components in this circuit in a nearly infinite number of ways, and as long as we made the same connections, then the circuit would work fine each time.

In our example, we connect the pins of each component right into the breadboard itself. If we had a more difficult circuit to build, we would need to use hookup wires to connect one point to another. These wires can either be solid, insulated wire cut to length, we find that 22-gauge works best, or male-to-male, pre-terminated jumper wires. Keep in mind though that when building a circuit on a breadboard, or changing some of the components or wiring, it's a good idea to disconnect any power source from the board, as crossing wires while the board is powered up could damage your components.

While the breadboard provides a quick and relatively painless way to prototype a circuit sometimes soldering two wires together, or parts like pin headers to breakout boards, is just unavoidable. So let's look at what soldering is and how we can make permanent connections with reasonable success.

Soldering

We love our solderless breadboards for making our lives that much easier, but occasionally there is no escaping the need to solder. Maybe the pins on a switch don't fit in the breadboard so we need to solder wires to it. Some of the breakout boards that we have been using do not already have pins attached when we buy them, so we need to solder pin headers to them to plug them into our breadboard. Figure 12-16 shows a couple of different breakout boards that will need a little soldering.

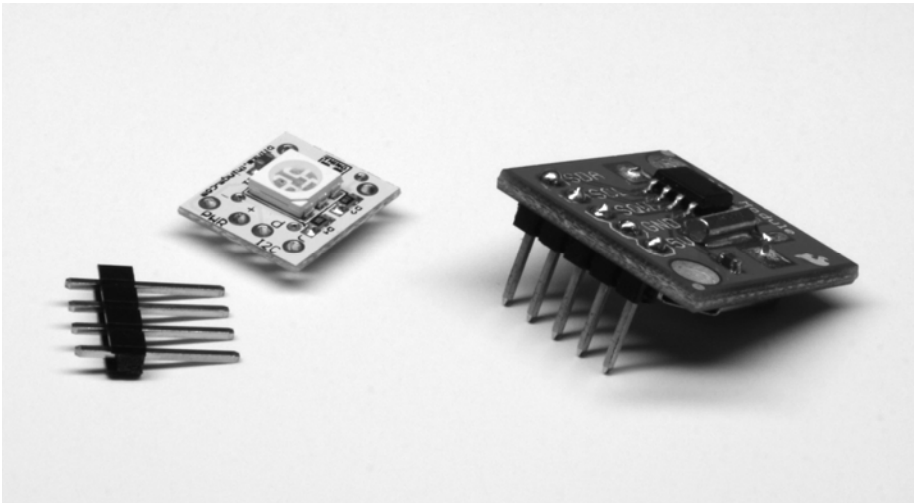


Figure 12-16. BlinkM MinM and SparkFun DS1307 RTC breakouts